

Debanjan Ray

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3rd year Computer Science student

Education

2023 - 2026

BSc Computer Science (1st), Newcastle University

Relevant Modules included: Major Project and Dissertation in Computer Science, Computer Vision and AI (84%), Development and Operation of Systems, System and Network Security (77%), Computer Systems Design and Architectures (85%), Foundations of Data Science (86%), Fundamentals of Computing (76%), Programming Portfolio 1 & 2 (82% & 92%), Security Programming, Algorithm Design and Analysis, Software Systems Design and Implementation.

- Designed and implemented an LLM-based research idea generation pipeline for an independent dissertation, pairing author publication histories retrieved from arXiv to investigate whether cross-domain context improves idea novelty. Evaluated outputs across five dimensions using Gemini 2.5 Flash, with statistical analysis (Mann-Whitney U, Spearman correlation) to test a domain-distance gradient hypothesis.
- Implemented PCA on the MNIST dataset to analyse and visualise key features of handwritten digits, demonstrating how dimensionality reduction captures dominant patterns and variability in image data.
- Developed a full-stack DevOps-focused web application using Flask and PostgreSQL, incorporating CI/CD pipelines, containerisation with Docker, and centralized logging via the ELK stack to enhance automation, scalability, and observability.
- Collaborated in a team of 5 to deliver a software implementation for generating and taking quizzes in Java. Discussed roles and tasks with other group members ensuring project deadlines were met. Developed over 700 lines of code, achieving a mark of 94.25%.
- Built part of a Relational Database based on storing and managing bus routes by formulating and constructing SQL queries in MySQL using DBeaver, achieving a mark of 94%.
- Developed a Reliable File Transfer protocol that works on top of UDP and guarantees reliable file delivery between a client and a server using network socket programming in Java.
- Created from scratch a web-based blog application using the Flask framework in Python allowing users to register accounts then create, edit and delete posts. Implemented multi-factor authentication and additional security features like rate limiting, access control and admin privileges.

A Levels: A (Maths), A (Biology), B (English Language)

9 IGCSEs: A* in all 9 subjects including Maths, Computer Science, Physics, Chemistry

Tech Skills

- Experienced in Python, C++, C#, and Java programming with experience using frameworks such as Flask and JUnit.
- Proficient in integrating Google Gemini API into complex projects.
- Experience using Google Colab for AI model training and visualisation using pandas library and Matplotlib.
- Knowledge of DevOps tools such as Docker, Slack, ELK Stack within a team
- Experience with using Git and Github collaboratively in a team.
- Knowledge of HTML and CSS for web design and styling.
- SQL, MySQL, PostgreSQL, have experience with DBeaver and MongoDB.
- Built a shell that can execute commands in the command line using the Go programming language.
- Proficient in C and Linux with Bash, using the command line.
- Developed games in Godot with GDScript.
- Proficient in Microsoft Office, including Outlook, Excel, Powerpoint and Word.

Achievements

Verizon Cloud Platform Job Simulation on Forage - November 2024

- Completed a job simulation involving building a hypothetical new VPN product for Verizon's Cloud Computing team.
- Used command line Python to test whether Verizon's VPN met the cloud-native traits, i.e. redundancy, resiliency and least-privilege.
- Researched approaches to achieve application security and communicated insights in a PowerPoint Presentation.

Achieved 1st place in the intermediate category of the writing competition 'In Search of Young Ruskin Bond'.

- The original short story was hand-chosen by the famous author Ruskin Bond. More than 4000 students from 84 schools across the country submitted entries.

Interests

- Executive committee member of the GameDev society, organises and participates in events, and connects with peers and industry professionals. Enjoys experimenting around in game engines such as Godot and Unity.
- Curious and passionate about learning new technologies, staying up-to-date through research and watching tutorials on different frameworks and tools.
- Regularly play chess online and is active in the chess society. Represented Newcastle university when playing at a friendly tournament with Durham University.